



UNSW  
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 5 – Determinants

Lecture 5.1 – Determinants in terms of permutations

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# Permutations

Recall that a **permutation** on  $n$  elements is an ordered arrangement of  $n$  distinct objects, or the action that converts one ordering to another. Recall also that the **symmetric group** of order  $n!$  is the group  $S_n$  of all such actions on  $n$  objects. For example, the permutation that converts the tuple  $(1, 2, 3)$  to  $(2, 1, 3)$  can be written as  $\sigma \in S_3$  where  $\sigma(1) = 2$ ,  $\sigma(2) = 1$ , and  $\sigma(3) = 3$ .

For a given  $\sigma \in S_n$ , a convenient way to encode the action of  $\sigma$  is to write it as the list  $[\sigma(1), \sigma(2), \dots, \sigma(n)]$ . So the permutation that swaps the first two of three ordered elements can be written as  $[2, 1, 3]$ .

**Definition.** For a given permutation  $\sigma \in S_n$ , an **inversion** is any pair  $i, j$  between 1 and  $n$  for which  $i < j$  but  $\sigma(i) > \sigma(j)$ .

For example, there is 1 inversion in the permutation  $[2, 1, 3]$ , and 4 inversions in the permutation  $[1, 4, 5, 2, 3]$  (for the pairs  $(2, 4)$ ,  $(3, 4)$ ,  $(2, 5)$ , and  $(3, 5)$ ).

**Definition.** The **sign** of a permutation  $\sigma$ , denoted  $\text{sign}(\sigma)$ , is given by  $\text{sign}(\sigma) = (-1)^k$  where  $k$  is the number of inversions in  $\sigma$ .

For example,  $\text{sign}([2, 1, 3]) = -1$  and  $\text{sign}([1, 4, 5, 2, 3]) = 1$ .

# Properties of permutation sign

**Lemma.** For any permutation  $\sigma \in S_n$ , we have

$$\text{sign}(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(i) - \sigma(j)}{i - j}.$$

**Proof.**

**Lemma.** For all  $\sigma, \tau \in S_n$ , we have  $\text{sign}(\sigma \circ \tau) = \text{sign}(\sigma) \text{sign}(\tau)$ .

**Proof.**

**Corollary.** The function  $\text{sign} : S_n \rightarrow \{-1, 1\}$  is a group homomorphism.

# Determinants

With the language of permutations, we can define the determinant of a square matrix in the following way.

**Definition.** The **determinant** of a matrix  $A \in M_{n,n}(\mathbb{F})$ , denoted  $\det(A)$  or  $|A|$ , is the scalar value

$$\det(A) = \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} = \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}$$

where  $a_{ij}$  is the  $(i,j)$ th entry of  $A$ .

We can check that this definition aligns with the one we are more familiar with from first year for some small cases:

**Example.** For the case  $n = 2$ , since  $S_2 = \{[1, 2], [2, 1]\}$ , we have

$$\begin{aligned} \det \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} &= \text{sign}([1, 2]) a_{1,1} a_{2,2} + \text{sign}([2, 1]) a_{1,2} a_{2,1} \\ &= a_{1,1} a_{2,2} - a_{1,2} a_{2,1}. \end{aligned}$$

## $3 \times 3$ determinants

**Example.** In the  $n = 3$  case, let the elements of  $S_3$  be  $\sigma_1 = [1, 2, 3]$ ,  $\sigma_2 = [1, 3, 2]$ ,  $\sigma_3 = [2, 1, 3]$ ,  $\sigma_4 = [2, 3, 1]$ ,  $\sigma_5 = [3, 1, 2]$ , and  $\sigma_6 = [3, 2, 1]$ .

Then for  $A = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix}$  we have by the permutation definition

$$\begin{aligned} \det(A) &= \text{sign}(\sigma_1)a_{1,1}a_{2,2}a_{3,3} + \text{sign}(\sigma_2)a_{1,1}a_{2,3}a_{3,2} + \text{sign}(\sigma_3)a_{1,2}a_{2,1}a_{3,3} \\ &\quad + \text{sign}(\sigma_4)a_{1,2}a_{2,3}a_{3,1} + \text{sign}(\sigma_5)a_{1,3}a_{2,1}a_{3,2} + \text{sign}(\sigma_6)a_{1,3}a_{2,2}a_{3,1} \\ &= a_{1,1}a_{2,2}a_{3,3} - a_{1,1}a_{2,3}a_{3,2} - a_{1,2}a_{2,1}a_{3,3} \\ &\quad + a_{1,2}a_{2,3}a_{3,1} + a_{1,3}a_{2,1}a_{3,2} - a_{1,3}a_{2,2}a_{3,1}. \end{aligned}$$

This aligns with the more familiar “expand along a row” method, for example:

$$\begin{aligned} \begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} &= a_{1,1} \begin{vmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{vmatrix} - a_{1,2} \begin{vmatrix} a_{2,1} & a_{2,3} \\ a_{3,1} & a_{3,3} \end{vmatrix} + a_{1,3} \begin{vmatrix} a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{vmatrix} \\ &= a_{1,1}a_{2,2}a_{3,3} - a_{1,1}a_{2,3}a_{3,2} - a_{1,2}a_{2,1}a_{3,3} \\ &\quad + a_{1,2}a_{2,3}a_{3,1} + a_{1,3}a_{2,1}a_{3,2} - a_{1,3}a_{2,2}a_{3,1}. \end{aligned}$$

# Properties of the determinant

**Lemma.** For any  $A \in M_{n,n}(\mathbb{F})$ , we have  $\det(A) = \det(A^T)$ .

**Proof.**

**Corollary.** For any  $A \in M_{n,n}(\mathbb{F})$ , we have  $\det(A^*) = \overline{\det(A^T)}$ .

(This is by linearity of the complex conjugate.)

# Properties of the determinant

**Lemma.** If the rows (or columns) of a matrix  $A \in M_{n,n}(\mathbb{F})$  are reordered by some permutation  $\pi \in S_n$  to create the matrix  $B$ , then  $\det(B) = \text{sign}(\pi) \det(A)$ .

**Proof.**

# Properties of the determinant

**Lemma.** If a matrix has two equal rows or columns, then its determinant is zero.

**Proof.**



# Properties of the determinant

**Lemma.** Given a matrix  $A \in M_{n,n}(\mathbb{F})$ , adding a multiple of a row/column to any other row/column does not change the determinant.

**Proof.**

# Multilinearity of the determinant

Notice that we can think of the determinant as a function on each of the columns of an  $n \times n$  matrix, that is, as  $\det : \underbrace{\mathbb{F}^n \times \cdots \times \mathbb{F}^n}_{n \text{ copies}} \rightarrow \mathbb{F}$ .

**Lemma.** When considered in this way, the determinant is an alternating multilinear map.

**Proof.**

# Matrix minors

**Definition.** Given a matrix  $A \in M_{n,n}(\mathbb{F})$ , the  $(i,j)$  minor of  $A$  is the matrix  $A_{i,j}$  obtained by removing the  $i$ th row and the  $j$ th column of  $A$ .

(Careful: similar notation is sometimes used for the  $(i,j)$ th entry of  $A$ .)

**Lemma.** In a matrix  $A \in M_{n,n}(\mathbb{F})$ , if the entry  $a_{i,j}$  is the only nonzero entry in the  $i$ th row, then  $\det(A) = (-1)^{i+j} a_{i,j} \det(A_{i,j})$ .

**Proof.**

(Note that this is a special case of the first-year “expand along a row” method.)

## Finding determinants

The previous result (which clearly also holds when “row” is replaced by “column”) can be used to help find determinants efficiently. Given any matrix, we can add multiples of rows/columns to other rows/columns until one of the rows/columns contains at most one nonzero entry. Then the formula from the previous lemma reduces the problem to finding the determinant of a smaller matrix, and this process can be repeated until the smaller matrix is  $1 \times 1$  (or just  $2 \times 2$ , since the formula for the determinant in that case is straightforward).

**Exercise.** Find the determinant of  $\begin{pmatrix} 2 & 0 & 2 & 0 \\ 3 & 2 & 4 & 2 \\ 4 & 1 & 2 & 2 \\ 5 & 2 & 3 & 1 \end{pmatrix}$ .

**Solution.**

# Finding determinants from diagonals

**Lemma.** The determinant of an upper (or lower) triangular matrix is given by the product of its diagonal entries.

**Proof.**

So another efficient method to find the determinant is to add multiples of rows/columns to other rows/columns until the matrix is upper or lower triangular, and then just take the product of the diagonals.

Another useful technique here is to note that swapping any two rows or columns has the effect of changing the sign of the determinant.

## Example

**Exercise.** By converting to an upper or lower triangular matrix, find the

determinant of the matrix  $\begin{pmatrix} 2 & 0 & 2 & 0 \\ 3 & 2 & 4 & 2 \\ 4 & 1 & 2 & 2 \\ 5 & 2 & 3 & 1 \end{pmatrix}$ .

**Solution.**